

Zack Andalman

PhD Candidate, Princeton University
751 Hibben Magie Rd, Unit 215, Princeton, NJ

zack.andalman@princeton.edu
zandalman.com

+1 847 208 5238
ORCID; ADS; ArXiv

EDUCATION

Princeton University	Princeton, NJ	Ph.D., Astrophysics	4.00 GPA	2023 - 2028
Yale University	New Haven, CT	B.S., Physics	3.95 GPA	2019 - 2023
Evanston Township HS	Evanston, IL	Diploma	4.00 GPA	2015 - 2019

PUBLICATIONS

First-Author

Published: 3

- [1] **Andalman, Z. L.**; Quataert, E.; Coughlin, E.; Nixon, C. (2026)
Resolving the (Debate About) Nozzle Shocks in Tidal Disruption Events. OJA.
<https://ui.adsabs.harvard.edu/abs/2025arXiv251208928A/abstract>
- [2] **Andalman, Z. L.**, Teyssier, R., & Avishai, D. (2025).
On the Origin of the High Star-Formation Efficiency in Massive Galaxies at Cosmic Dawn. MNRAS.
<https://ui.adsabs.harvard.edu/abs/2024arXiv241020530A/abstract>
- [3] **Andalman, Z. L.**; Liska, M. T. P.; Tchekhovskoy, A.; Coughlin, E. R.; & Stone, N. (2022).
Tidal Disruption Discs Formed and Fed by Stream-stream and Stream-disc Interactions in Global GRHD Simulations. MNRAS.
<https://ui.adsabs.harvard.edu/abs/2022MNRAS.510.1627A/abstract>

Submitted: 1

- [4] **Andalman, Z. L.**; Fryer, C.; Fontes, C.; Mumpower, M.; Wolleager, R. (2026).
Beta-Particle Transport and Thermalization in Kilonova Ejecta with Detailed Atomic Microphysics. MNRAS.

Co-Author

Published: 2

- [5] Sunseri, J.; **Andalman, Z. L.**; Teyssier, R. (2026).
Supermassive Black Hole Growth in Massive Galaxies at Cosmic Dawn. MNRAS.
<https://ui.adsabs.harvard.edu/abs/2025arXiv251019822S/abstract>
- [6] Kaaz, N.; et. al. incl. **Andalman, Z. L.** (2023).
Nozzle Shocks, Disk Tearing and Streamers Drive Rapid Accretion in 3D GRMHD Simulations of Warped Thin Disks. MNRAS.
<https://ui.adsabs.harvard.edu/abs/2023ApJ...955...72K/abstract>

Submitted: 1

- [7] Nixon, C.; Coughlin, E.; & **Andalman, Z. L.** (2026).
On the Origin of Anomalous Dissipation in Simulations of Tidal Disruption Events. ApJ.
<https://ui.adsabs.harvard.edu/abs/2026arXiv260520327N/abstract>

PRESENTATIONS

Selected Talks

TDEs and Nuclear Transients in Crete	2026
Inaugural Tinsley workshop	2024
LANL Center for Nonlinear Studies Student Talk Series (1st place student talk)	2024
KITP Program - Towards a Physical Understanding of Tidal Disruption Events	2024
RAMSES User Meeting	2024
HEAD Frontiers Seminar Series	2023
HEAD-19 Conference (invited)	2022

Selected Posters

AAS-245 Conference (Chambliss honorable mention)	2025
HEAD-20 Conference (1st place undergraduate poster)	2023
Connecticut Space Grant Consortium Expo	2021, 2022
Blue Waters Symposium for Petascale Science and Beyond	2018, 2019

SKILLS

Computer languages: PYTHON, C, C++, FORTRAN, HTML/CSS/JAVASCRIPT, Unix shell

Software: H-AMR, RAMSES, ATHENA++/K, CLOUDY, MESA

Languages: English (native), Spanish (conversational), Modern Greek (basic)

Hobbies: Triathlon, jazz piano

GRANTS, FELLOWSHIPS, AND AWARDS

DOE Computational Science Graduate Fellowship, Krell Institute	2023 - 2027
Martin Schwarzschild Fellowship, Princeton University (departmental award)	2023 - 2025
Michael Manzella Award, Yale University (leadership award)	2023
Lamat Fellowship, University of California Santa Cruz (REU)	2022
Hahn Scholarship, Yale University (competitive undergrad research program)	2019 - 2021
Student Project Grant, NASA Connecticut Space Grant Consortium	2020
First-Year Summer Research Fellowship, Yale University	2020

LEADERSHIP EXPERIENCE

Thunch (thursday lunch) seminar series organizer	2024
Yale Undergraduate Aerospace Association; Project Leader, Director of Projects, President	2020 - 2023
Yale Club Triathlon, Captain	2021 - 2022

PROFESSIONAL SERVICE

Number of papers refereed

MNRAS: 2

YouTube channel with scientific visualizations

<https://www.youtube.com>

TEACHING, OUTREACH, AND SCIENCE COMMUNICATION

Recurring

Teaching fellow, Princeton University Preparatory Program	2025 - 2027
Mentored undergraduate researcher for summer research program	2026
Teaching assistant, AST 206: Black Holes	2024 - 2026
Tutor, New Jersey Prison Teaching Initiative	2024
Organizer, multiple aerospace-themed educational events with local schools	2021

One-time

Scientific visualization featured in Quanta magazine <i>Astrophysicists Puzzle Over Webb's New Universe</i> www.quantamagazine.org/...20260702	7/22, 2026
Volunteer, Princeton Spring into Science (educational event with local schools)	4/16, 2026
Speaker, Planetarium at New Jersey State Museum	2/21, 2026
Speaker, Astronomy on Tap (Trenton, NJ)	11/20, 2024
Speaker, solar eclipse at Littlebrook Elementary	4/08, 2024
Teacher, Yale Splash (annual event with local schools)	4/16, 2022